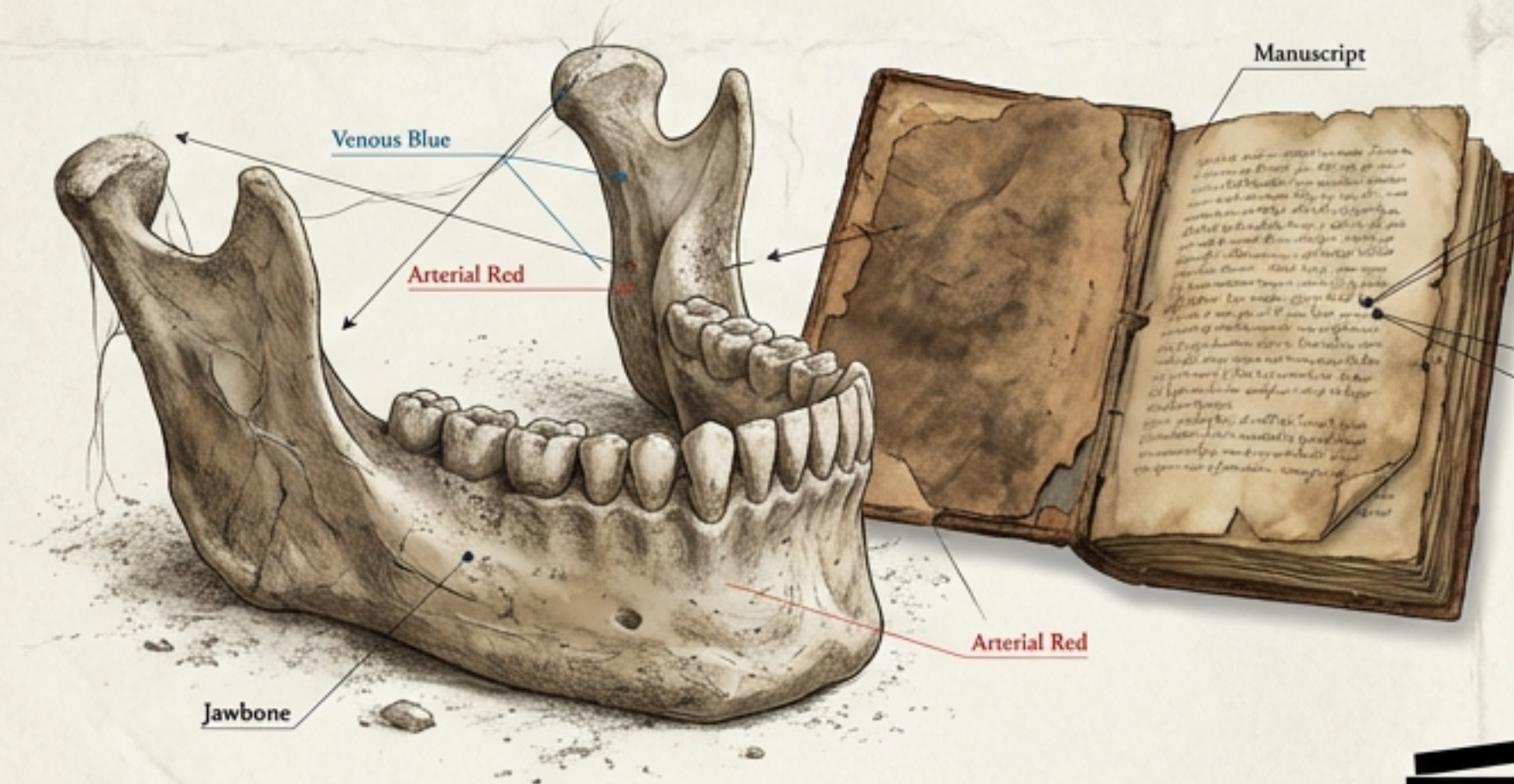


The Human Story Retold Through Our Genes

A biological detective story about everyone who ever lived.

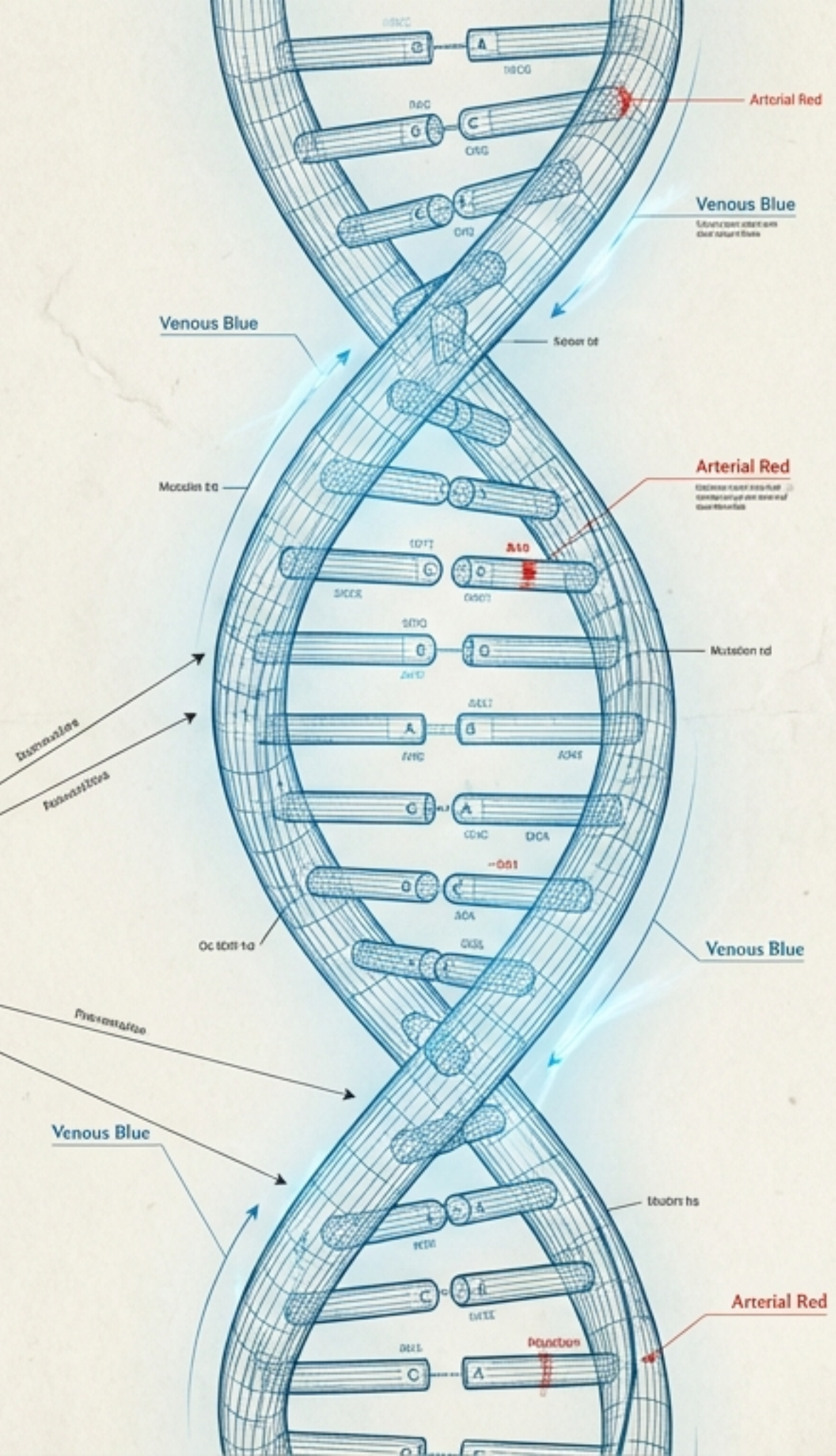
History is what we recorded. Prehistory is what happened before we wrote it down. But you carry an epic poem in your cells: 3 billion letters of DNA that predate all written language.

For centuries, we relied on stones and bones to guess where we came from. Now, we have a new source. We have learned to read the genome, and it is rewriting our past, correcting our myths, and solving the oldest mysteries of our species.

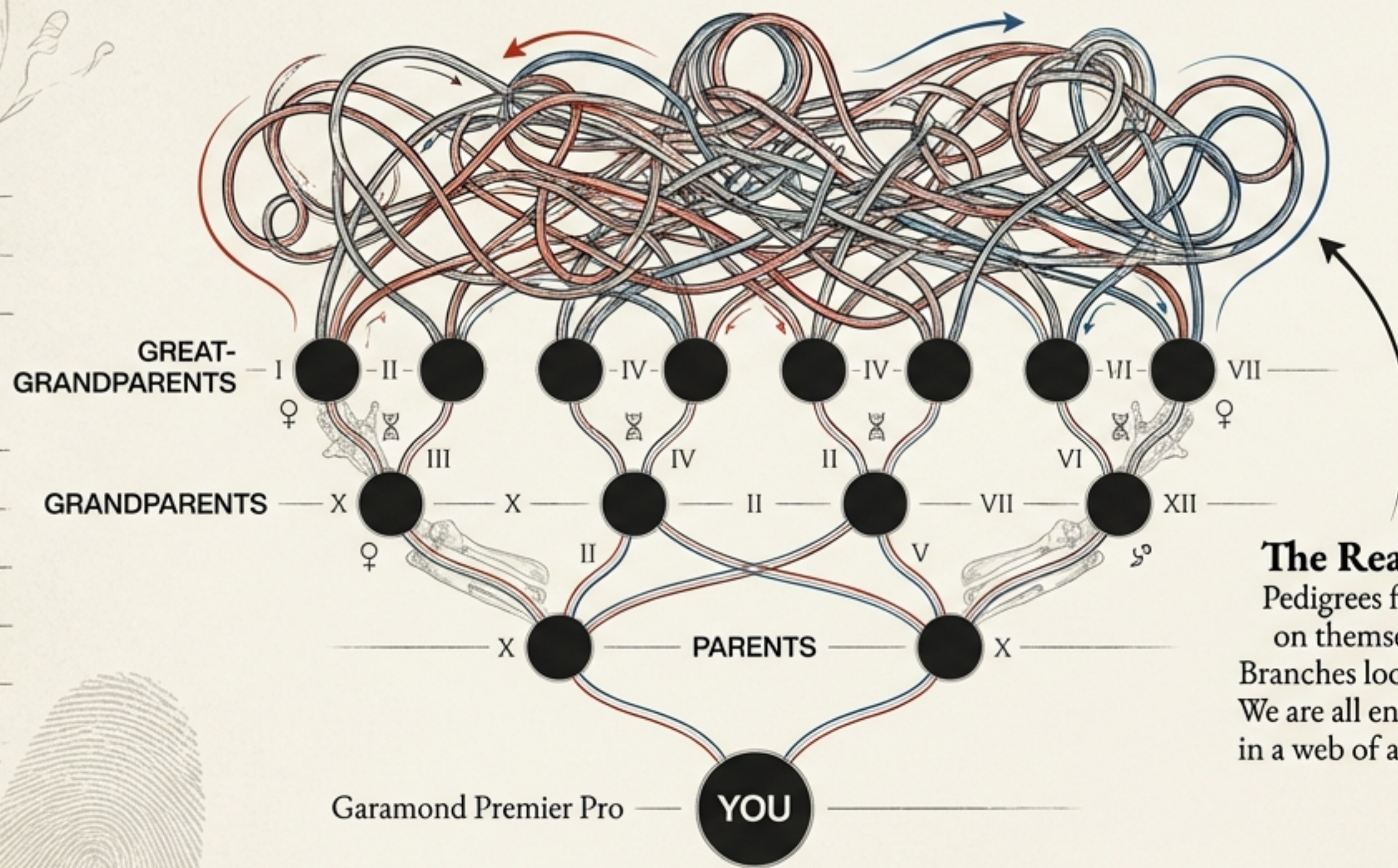


Based on the work of Adam Rutherford.

THE EXPERIMENT
NEW YORK



Your Family Tree Is Not a Tree



The Reality:
Pedigrees fold in
on themselves.
Branches loop back.
We are all enmeshed
in a web of ancestry.

The Math of Ancestry:

Generation 1:
2 parents

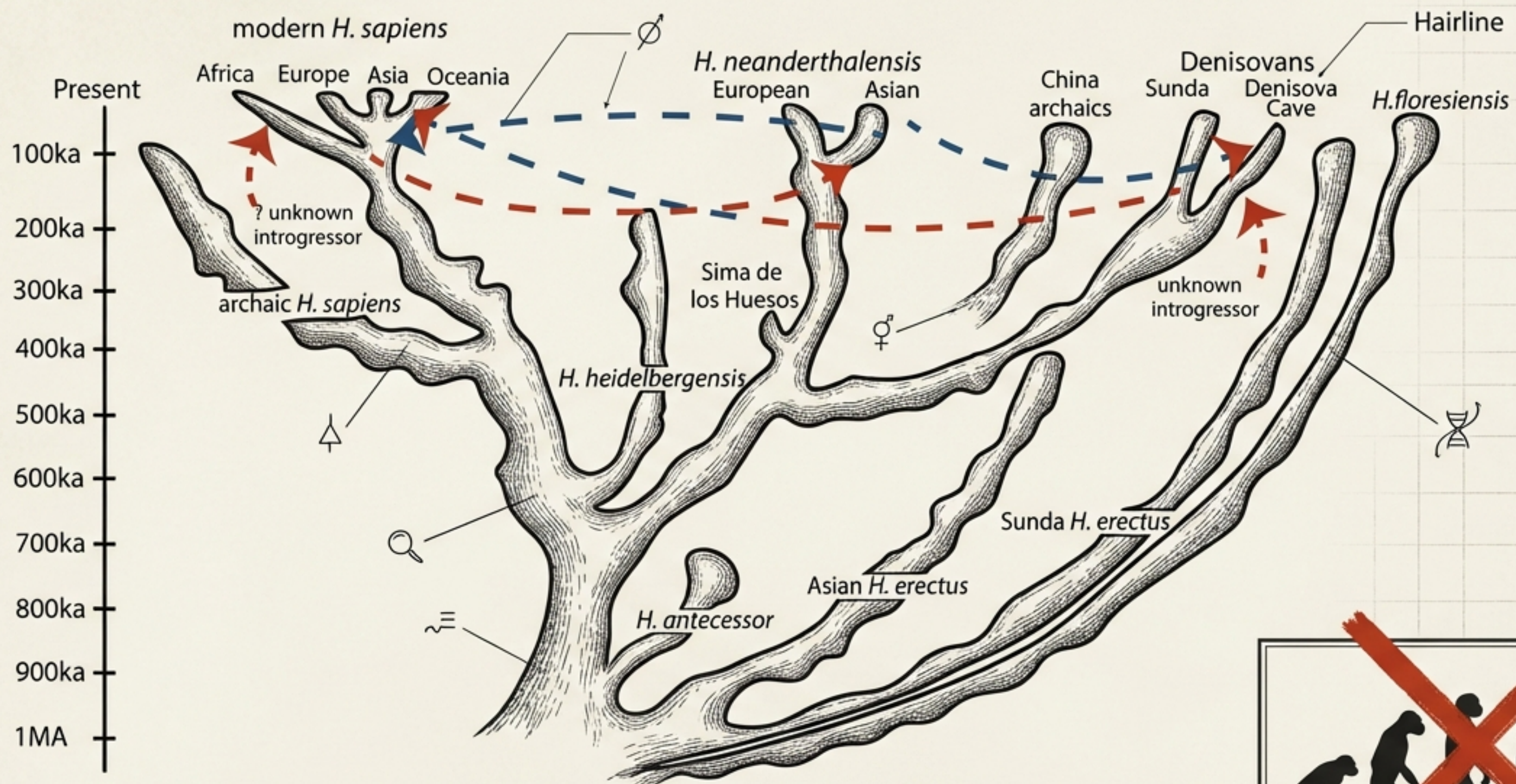
Generation 10:
1,024 ancestors

Generation 30:
~1 billion ancestors
(Mathematical impossibility)

Generation 40:
~1 trillion ancestors
(More than have ever lived)

We are 107 billion cousins, descended from a tiny handful of people, entwined in a mesh of biology and fate.

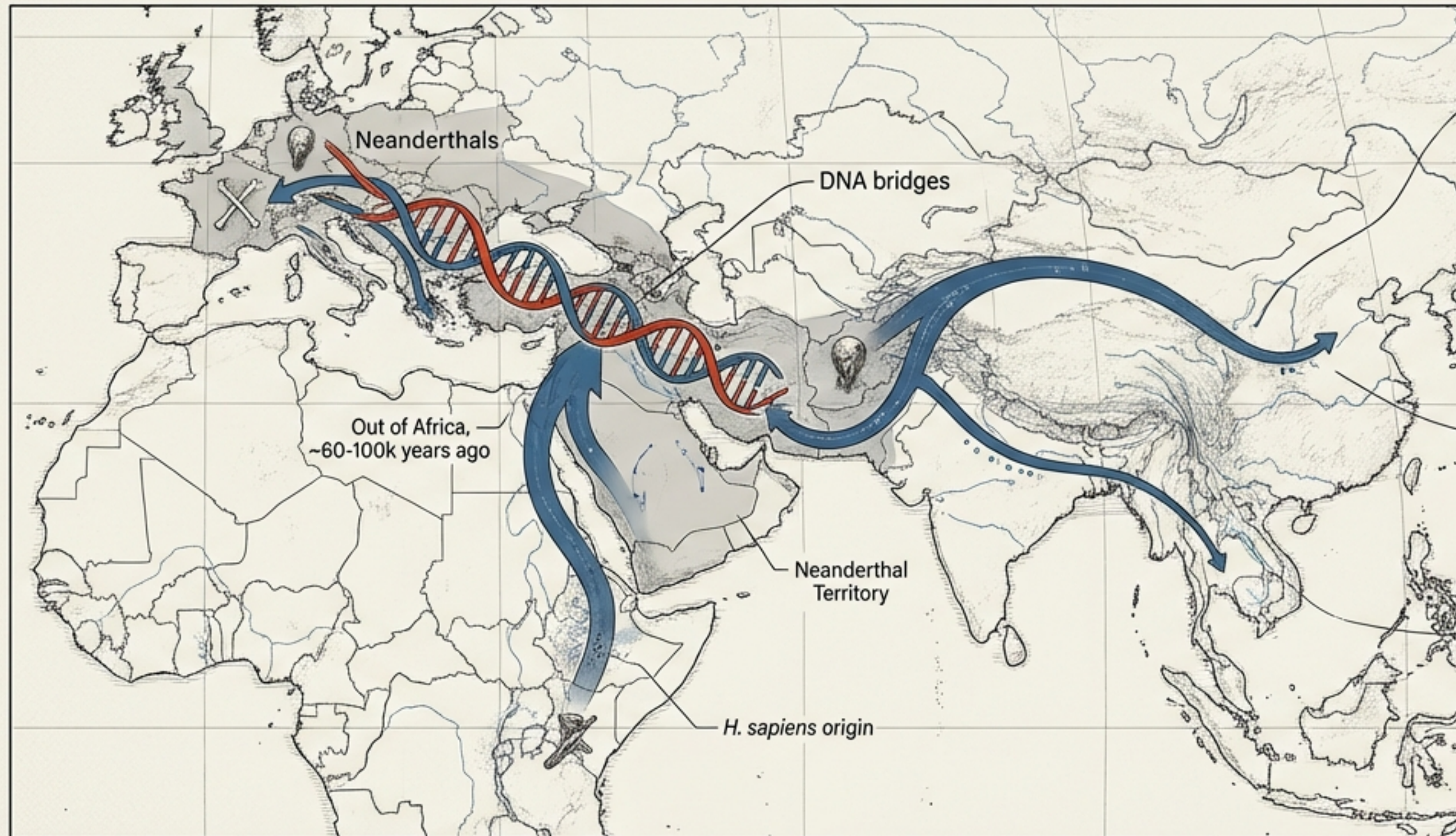
The Million-Year Clusterf*ck



The Old View: A neat line of progression. Monkey → Ape → Caveman → Us.
The Genetic Reality: A pollarded, tortuous shrub. **The Insight:** There are no missing links. Just the ebb and flow of living through epochs. Evolution doesn't march; it stumbles, branches, and merges.



Horny and Mobile



The Event:

When *Homo sapiens* broke out of Africa (approx. 60,000–100,000 years ago), we didn't just meet our cousins, the Neanderthals. We mated with them.

The Evidence:

2.7% of the DNA in modern Eurasians comes from Neanderthals.

The Verdict:

Neanderthals were not just our cousins; they were our ancestors.

Migration routes and genetic interactions visualised.
Data reflects scientific understanding of ancient human movements and admixture.

Rehabilitating the “Caveman”

The Stereotype



Outdated perception.

The Reality



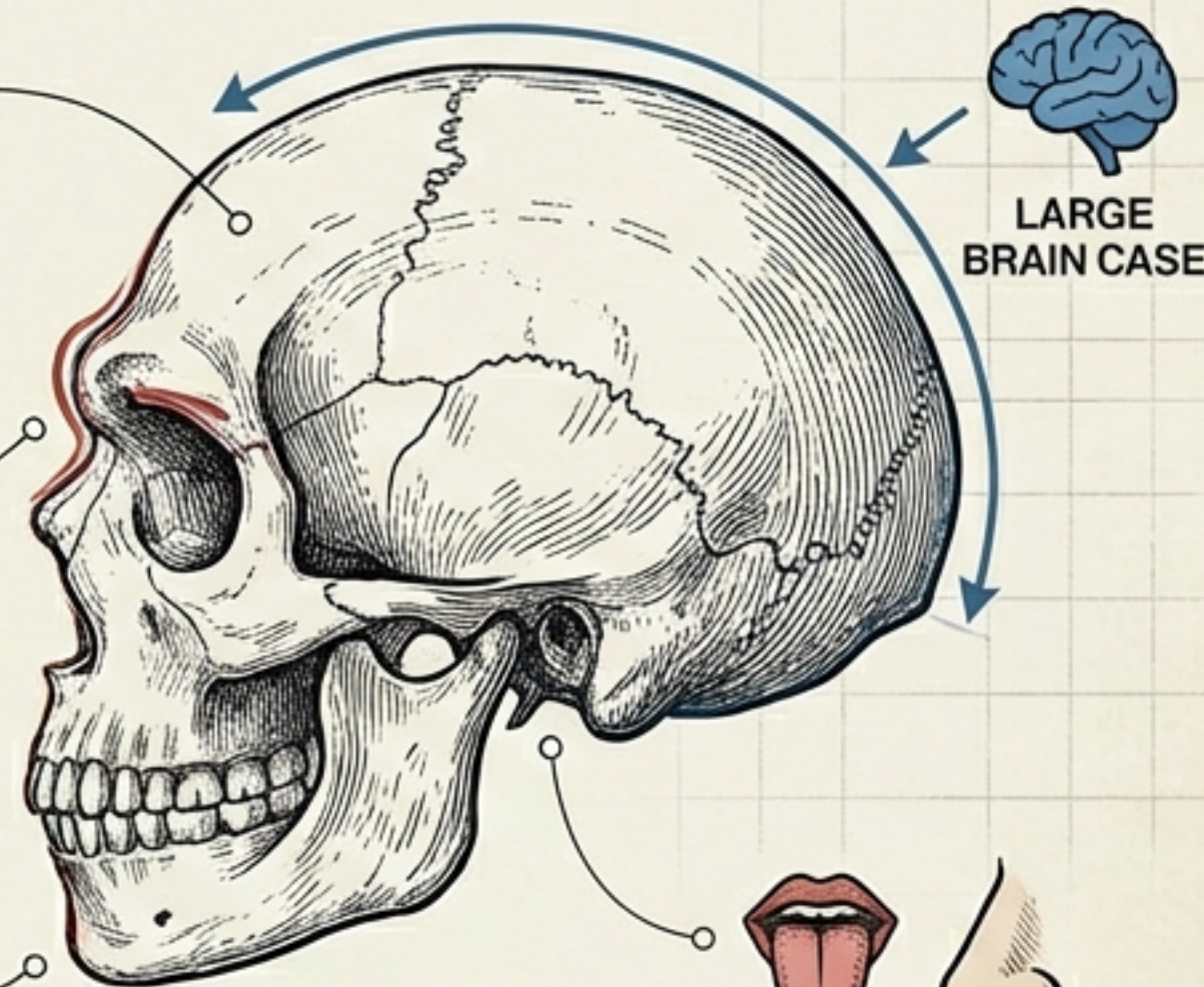
Brain Size: 1.2–1.7 liters
(Larger than modern human average).



Speech: FOXP2 gene & hyoid bone structure suggest full speech capability.



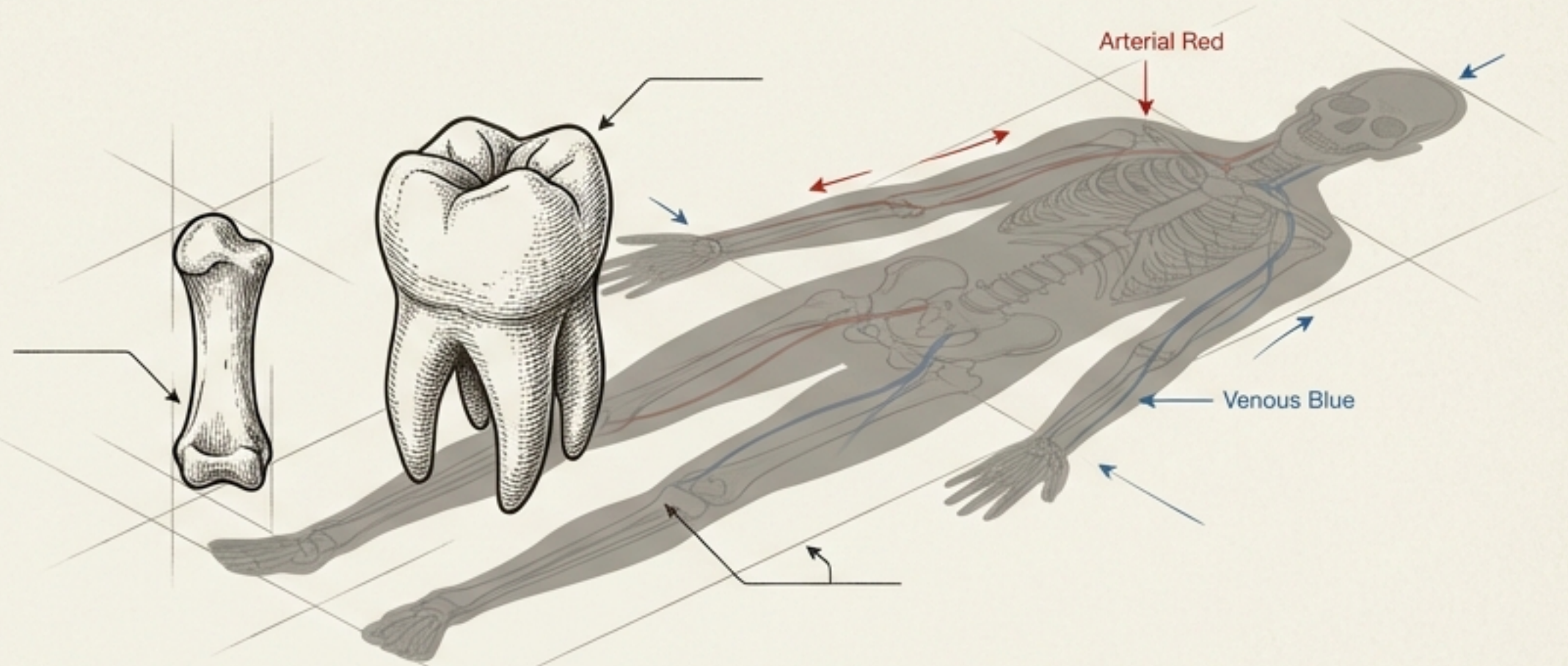
Culture: Jewelry, cooking, cooking, potential cave art.



LARGE BRAIN CASE

Senses: DNA reveals distinct smell receptors and ability to taste bitter flavors.

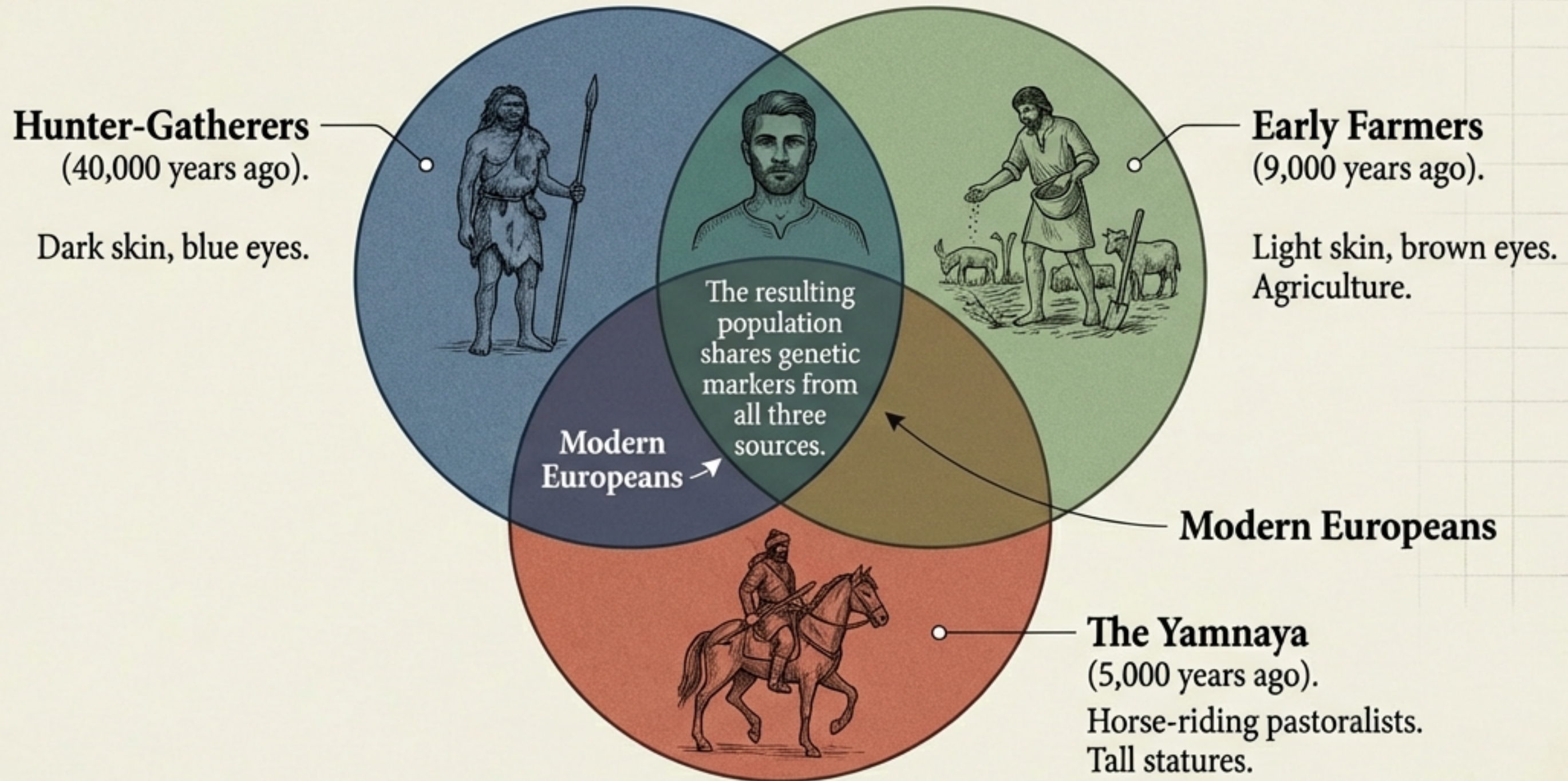
The Ghost in the Machine: The Denisovans



- ❧ **The Clue:** A single pinky bone and a molar found in a Siberian cave (2008).
- ❧ **The Revelation:** DNA sequencing revealed a completely new species of human—neither *Homo sapiens* nor Neanderthal.
- ❧ **The Legacy:** Their DNA lives on in Melanesians and Tibetans (providing high-altitude survival genes).
- ❧ **The Mystery:** Genetic analysis suggests the Denisovans bred with yet another unknown “ghost” species.

The First European Union

Modern Europeans are a blend of three distinct ancestral lineages.



There is no “indigenous” European bloodline that has remained pure. We are a mix of waves colliding.

The Mystery of the Milk Drinkers

Gene-Culture Coevolution

The Norm: Most adult humans cannot digest milk (lactose intolerance is the default).

The Mutation: Around 7,500 years ago, a mutation in the LCT gene appeared here.

Gene-Culture Coevolution: We started dairy farming *before* we could digest milk.
The cultural practice drove the biological evolution.

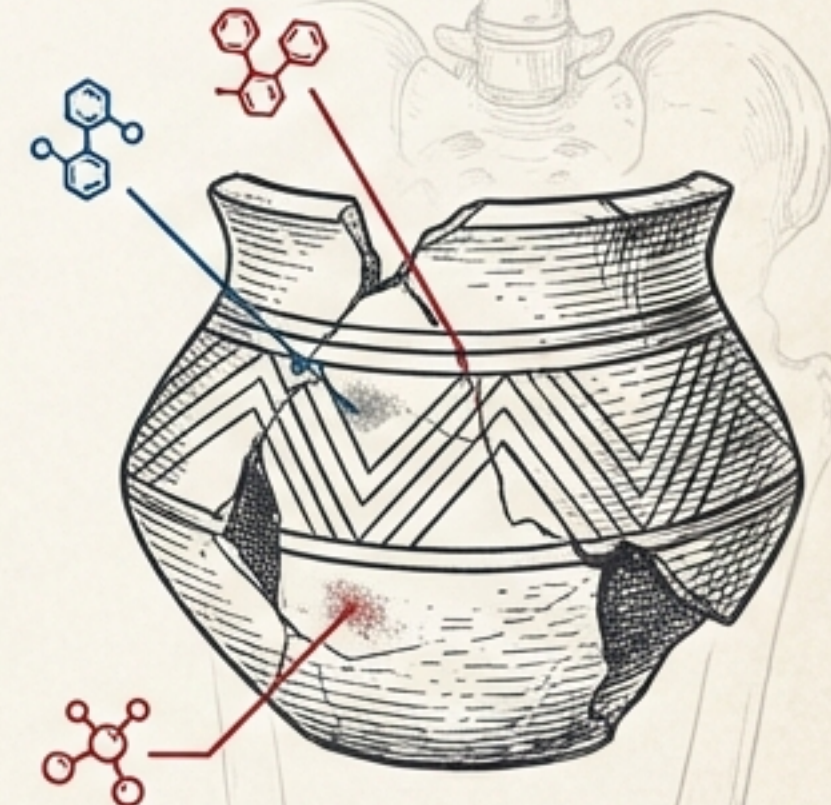


LCT Gene Mutation
(lactase persistence)

Lactase
Persistence

Increased
Survival

Dairy Diet
Adaptation



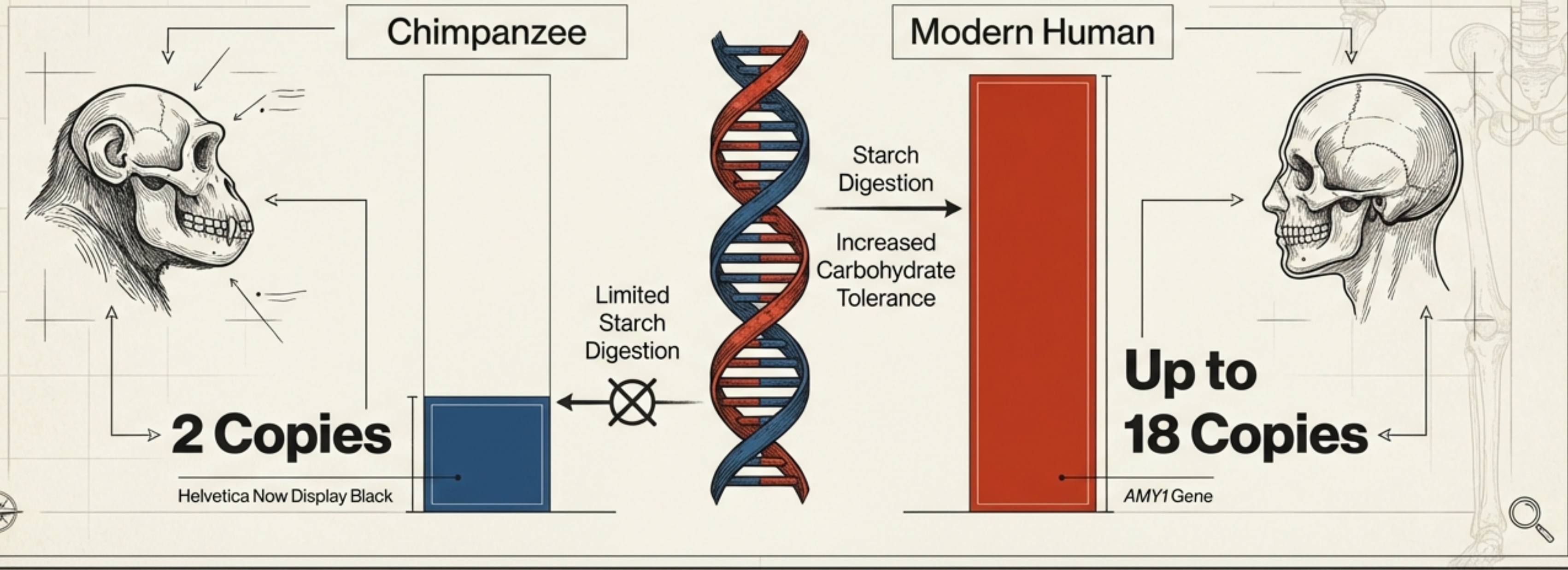
Milk Fat Residue Analysis
(Lipid Biomarkers)

Why the “Paleo Diet” is Bunk

The Myth: We haven’t evolved since the Paleolithic era.

The Science: We have evolved massively. The expansion of the Amylase gene (*AMY1*) allows us to digest starch and cooked carbohydrates—foods the Paleo diet often forbids.

The Reality: Go ahead and diet, but don’t pretend it’s based on evolutionary precedent.



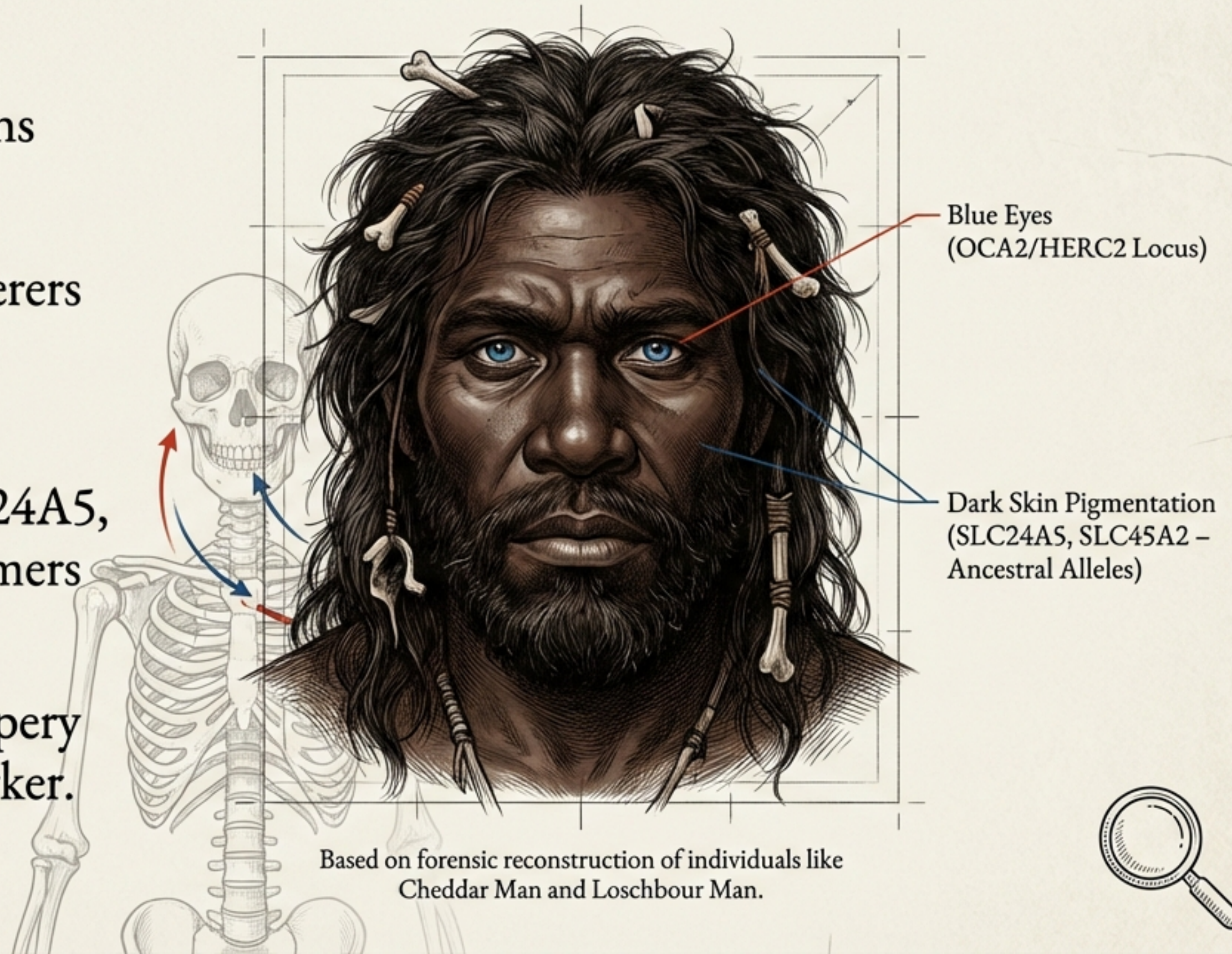
The Recent Invention of 'White People'

The Old Assumption: Europeans have always been pale.

The Genetic Fact: Hunter-gatherers in Europe 8,000 years ago had dark skin and blue eyes.

The Shift: Pale skin genes (SLC24A5, SLC45A2) arrived later with farmers and the Yamnaya.

The Lesson: Skin color is a “slippery spectrum,” not a fixed racial marker.



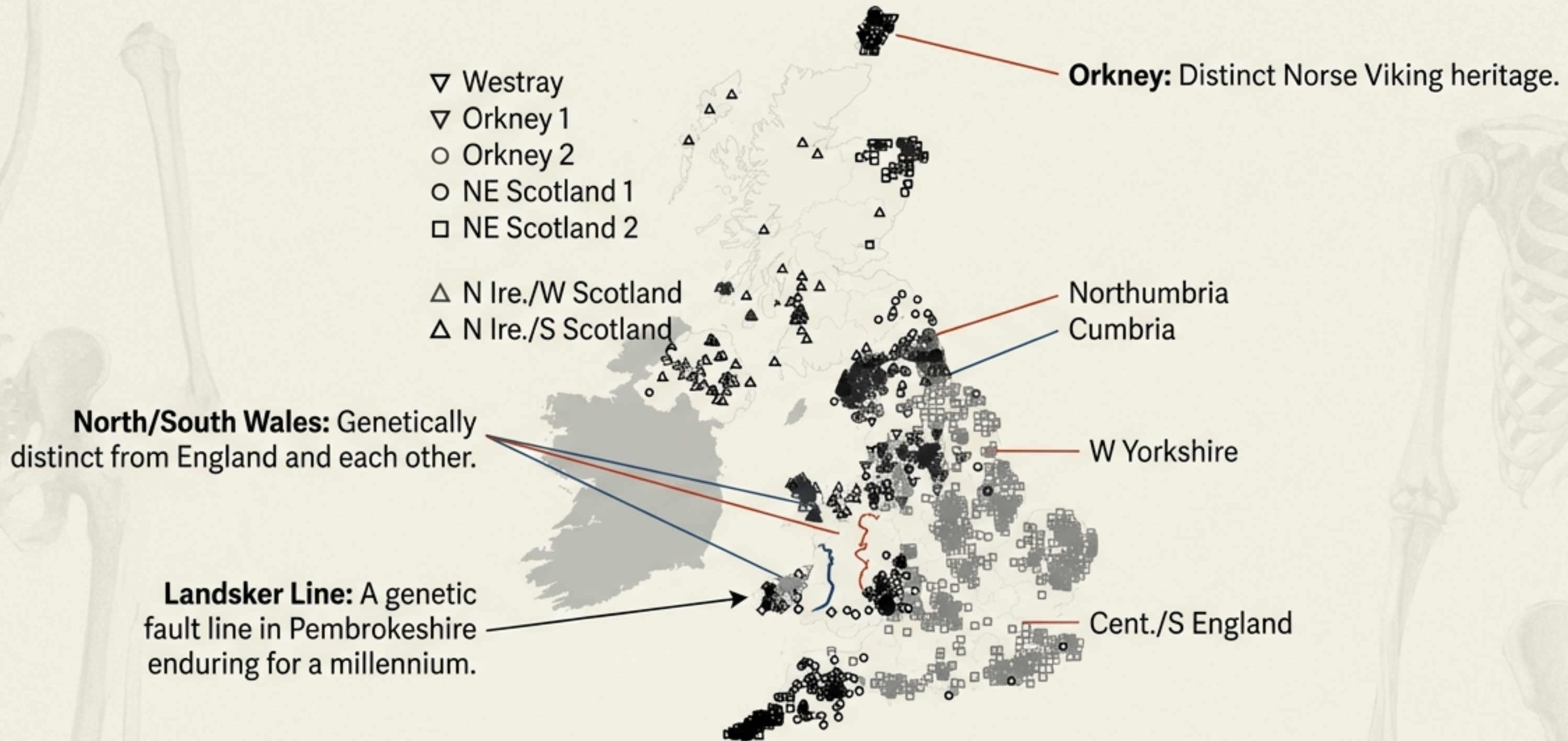
Blue Eyes
(OCA2/HERC2 Locus)

Dark Skin Pigmentation
(SLC24A5, SLC45A2 –
Ancestral Alleles)

Based on forensic reconstruction of individuals like
Cheddar Man and Loschbour Man.



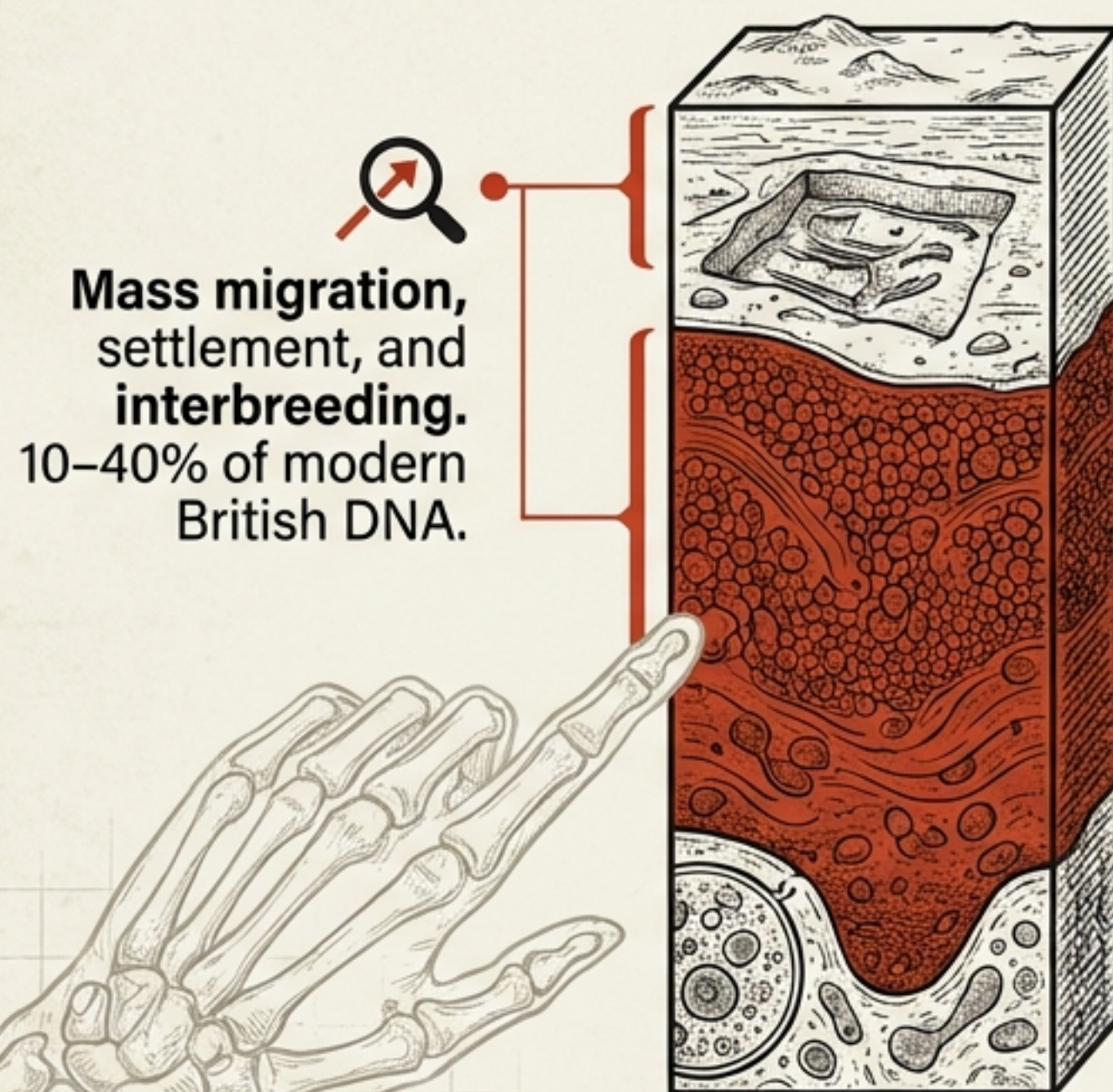
A Genetic Map of the British Isles



Based on the 'People of the British Isles' study. Genetic clusters align perfectly with 6th-century kingdoms and geographical boundaries.

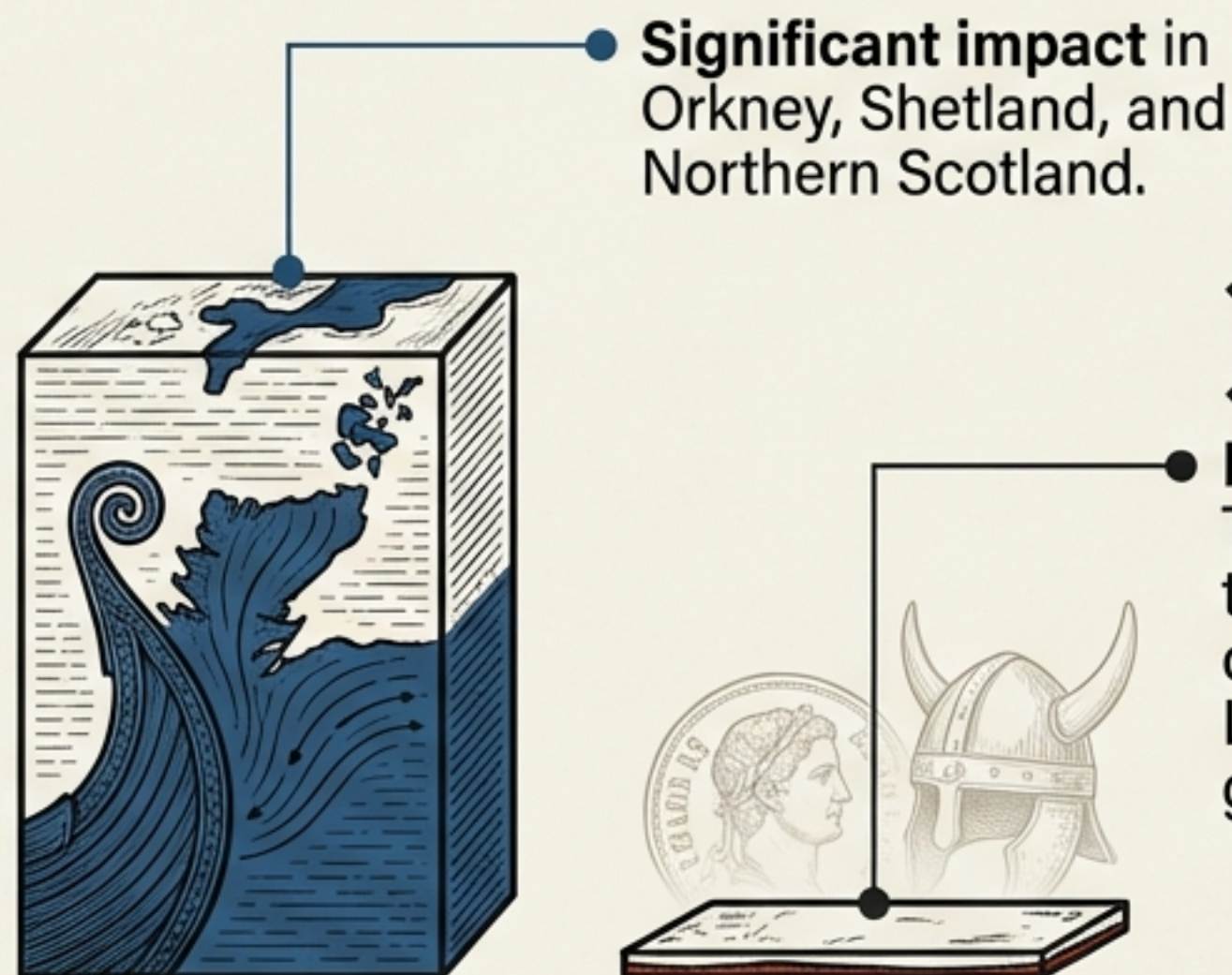
Who Are the British? (And Who Aren't They?)

Genetic Impact of Invasions on Britain



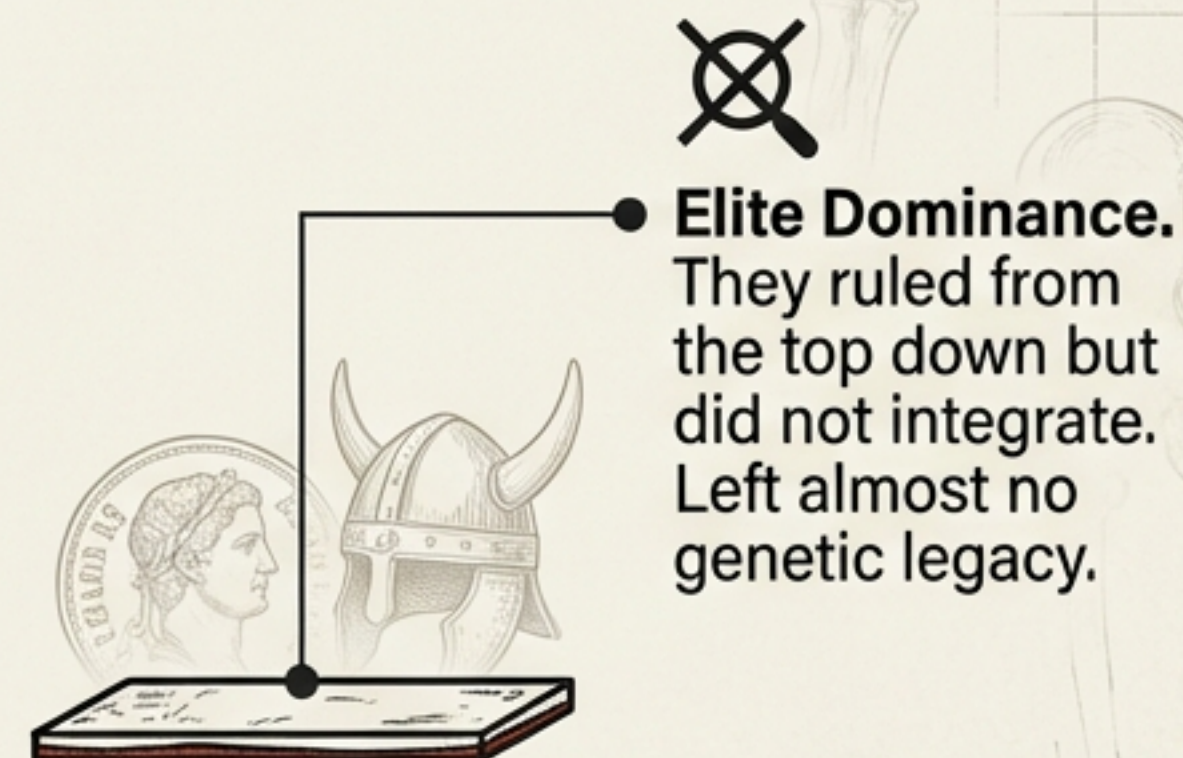
Mass migration, settlement, and interbreeding.
10–40% of modern British DNA.

Anglo-Saxons



Significant impact in Orkney, Shetland, and Northern Scotland.

Norwegian Vikings



Elite Dominance.
They ruled from the top down but did not integrate. Left almost no genetic legacy.

Romans & Danish Vikings

The Icelandic Experiment

Founder Effect

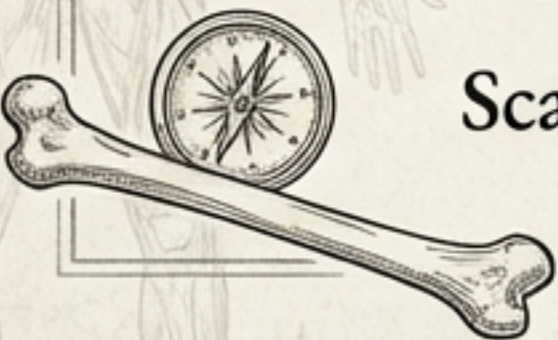
The History: Settled in the 9th century by Norse men and Celtic women.

Genetic Drift: Because the founding population was so small, Icelanders drifted genetically away from the mainland.

The Result: A unique genetic laboratory where recessive traits and “loss of function” mutations are common.

Scandinavia & British Isles

Iceland



GINGERS FACE EXTINCTION!

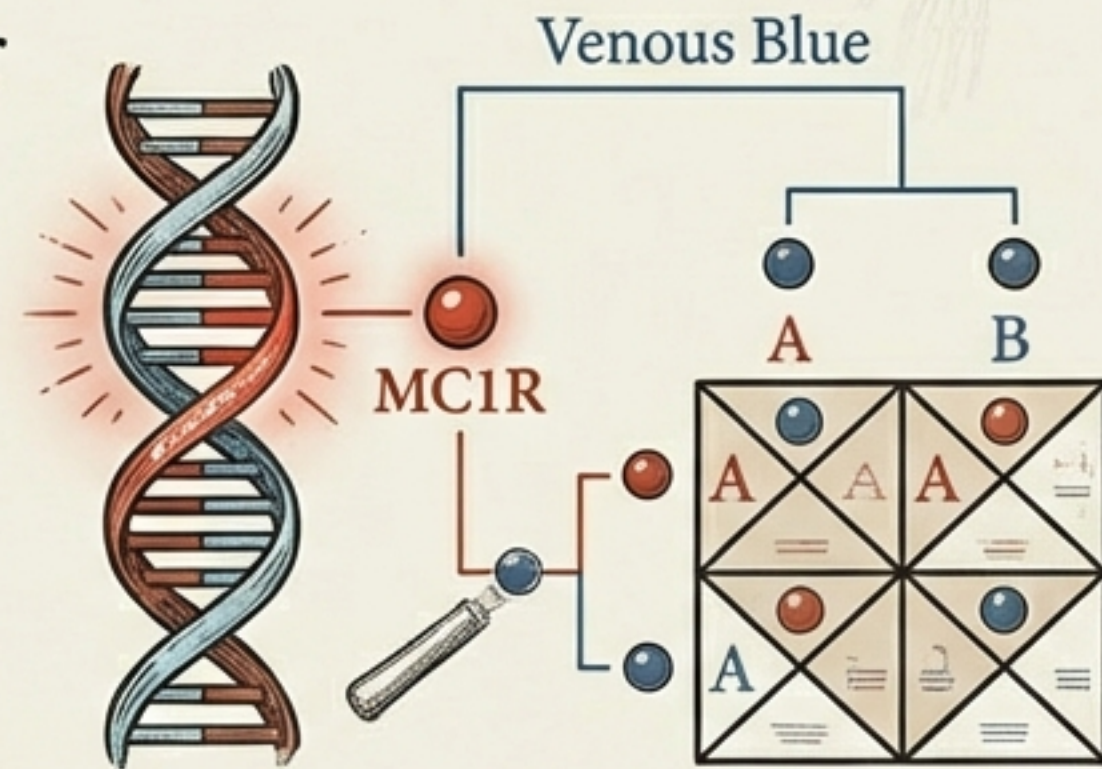
FALSE

The Ginger Extinction Hoax

The Scare: Media reports claimed sunnier weather would eliminate the need for red hair.

The Science: Red hair (MC1R gene) is recessive. It hides in the genome even if not expressed.

Author's Note: "Redheads aren't going anywhere unless we all stop having sex." Genes don't disappear just because they aren't currently useful.

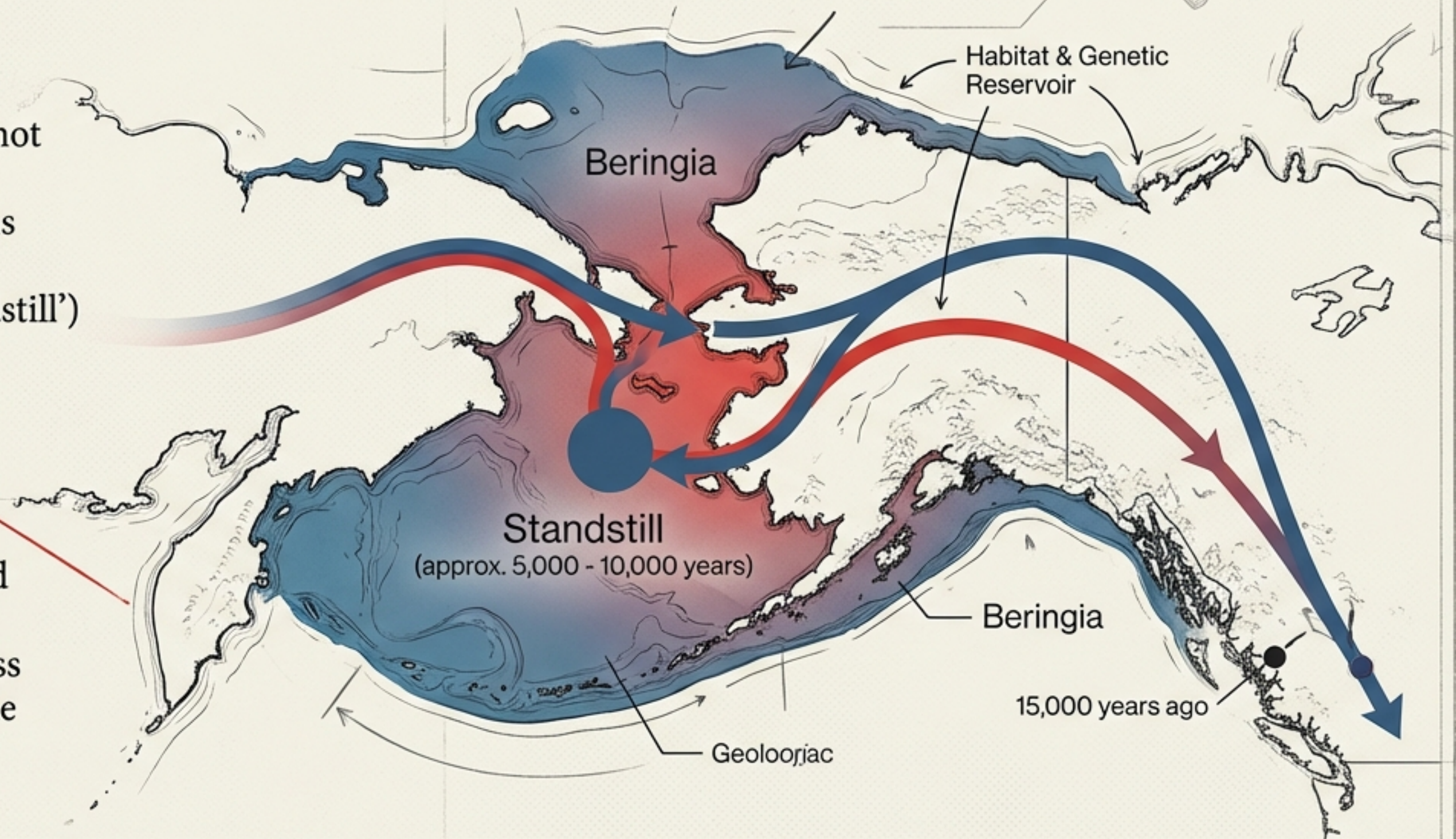


The Beringian Standstill

The New View: Beringia was not just a bridge; it was a habitat. Ancestors of Native Americans lived and evolved there for thousands of years (the 'Standstill') before moving south.



The Tragedy: Colonialism and genocide have left a "Black Hole" in our data. We know less about the genetic history of the Americas than we should.



The Invisible Sculptor

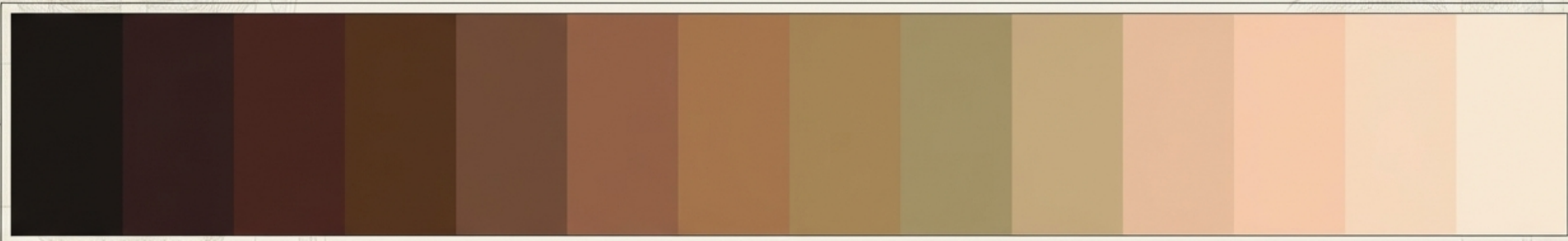
The Killer: *Yersinia pestis*
(The Black Death). 💀

The Impact: Killed 30-50%
of Europe. 🧬

The Scar: We see the battle scars in our DNA today. European populations show “selective sweeps” in immune system genes (TLR genes) absent in populations that didn’t face the plague. 💀



The End of Race



The Science: Genetic diversity *within* any “race” is often greater than the diversity *between* races.



The Verdict: Race is a social construct, not a biological one.

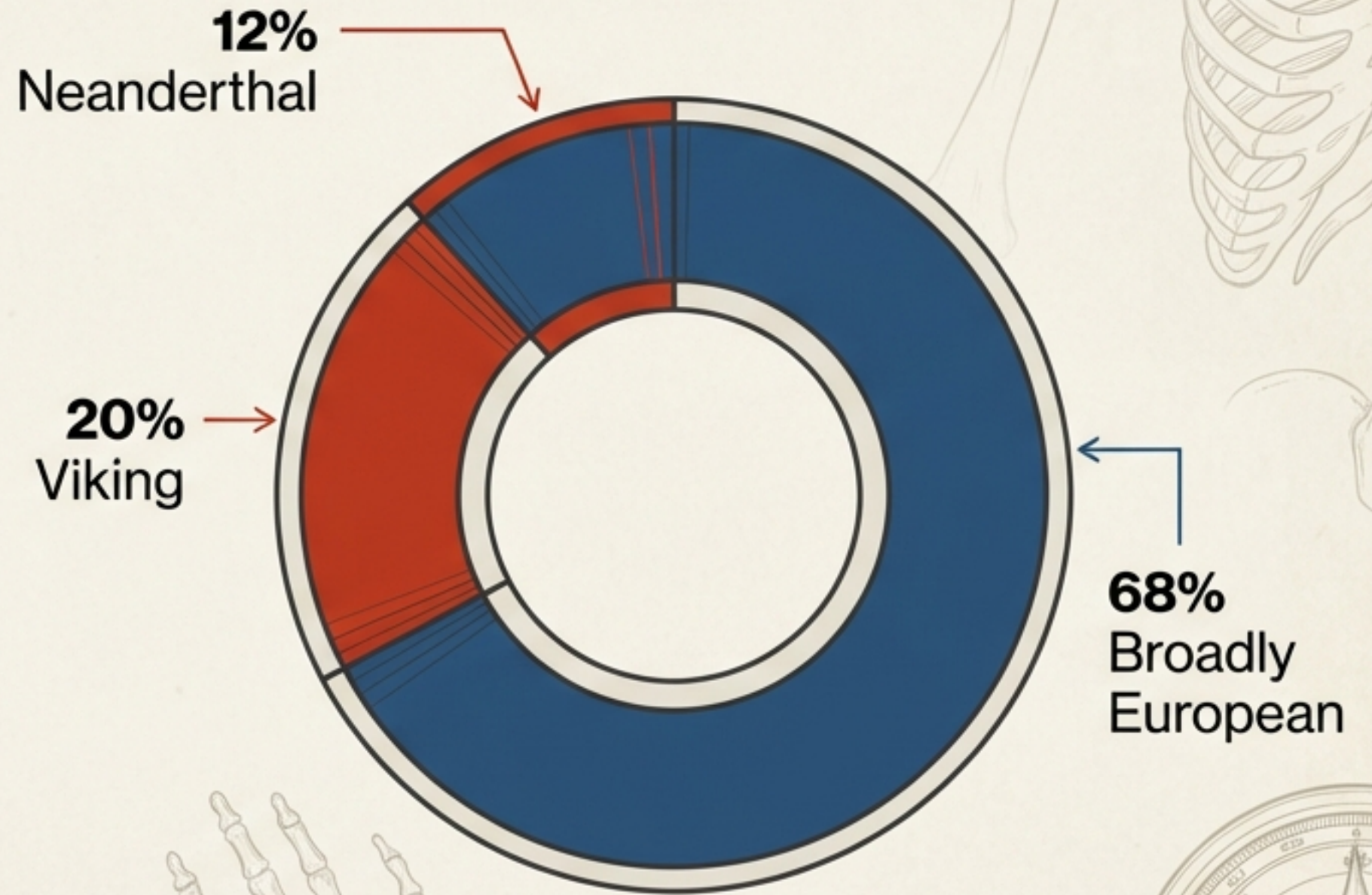
“ “There are no purebred humans. We are all mongrels, enriched by the genes of the multitudes who came before us.” ”

Astrology or Science?

The Promise: Telling you exactly who you are.

The Reality: It is probability, not destiny. It tells you about your DNA's similarity to current populations, not necessarily ancient ones.

The Warning: Genetics won't tell you how smart your kids will be, or what sports they should play. It tells you about your history, not your fate.



You Are Descended from Royalty (And Thieves)

If you are European, you are descended from Charlemagne.
You are also descended from his servants.

The Universal Truth: Go back 1,000 years, and everyone alive then who has descendants today is the ancestor of *everyone* alive today.

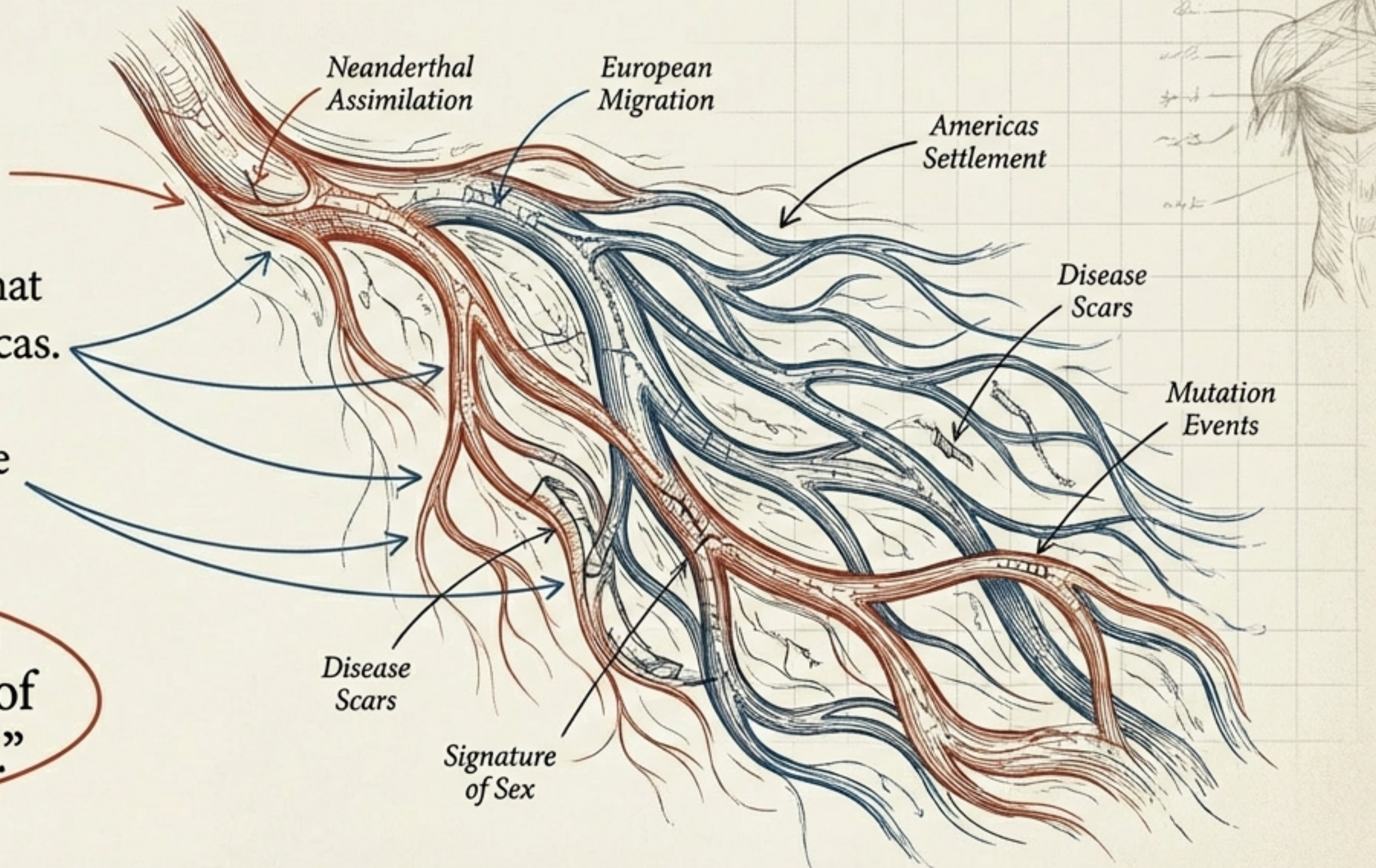
“Genealogy is narcissistic;
genetics is democratic.”



The River, Not the Tree

- We solved the case of the Neanderthals (assimilation).
- We tracked the migrations that made Europe and the Americas.
- We found the scars of disease and the signatures of sex.

“The past is not a foreign country; it is a sequence of letters written inside you.”



A Brief History of Everyone Who Ever Lived



You are the result of 4 billion years of evolutionary success.
An unbroken chain of survivors. DNA is a record of the journey life has taken to get to you.
It is the only form of immortality we have.

Based on the book by Adam Rutherford.